

BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

End-to-End Data Engineering, Quantitative Financial Analytics, Risk Modeling
& Power BI Executive Dashboard

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Project Domain:	Indian Mutual Fund Industry Analytics & Investment Research
GitHub Repository:	github.com/DGskywalker/BlueStock
Release Version:	v1.0 Capstone Release
Date of Execution:	August 2026

Executive Brief: This capstone project delivers an end-to-end analytics platform for the Indian Mutual Fund industry. Integrating 10 comprehensive datasets spanning daily NAV histories (64,320 records), fund master metadata, AUM trends (₹81.4 Lakh Cr), investor transaction logs (32,778 records), and portfolio stock disclosures, the platform automates data ingestion, cleaning, SQLite star schema storage (bluestock_mf.db), quantitative risk diagnostics (CAGR, Sharpe, Sortino, OLS Alpha/Beta, 95% VaR & CVaR, HHI concentration), and presents interactive insights via a 4-page Power BI executive dashboard.

1. Executive Summary & Project Background

The Indian Mutual Fund industry has experienced phenomenal growth over the last decade, with total Assets Under Management (AUM) surpassing ₹81.4 Lakh Crores and retail Systemic Investment Plan (SIP) monthly inflows reaching an all-time peak of ₹31,002 Crores in December 2025. However, institutional investors, wealth managers, and retail participants face significant data fragmentation, unstandardized transactional records, and a lack of unified risk-adjusted metrics.

Key Capstone Objectives Achieved:

- Production Data Ingestion & ETL Pipeline:** Successfully ingested 10 cleaned CSV datasets, validated AMFI codes against daily NAV histories with 100% integrity, and built automated forward-fill mechanisms for holiday/weekend NAV gaps.
- SQLite Star Schema Database (bluestock_mf.db):** Designed and loaded an 11-table relational star schema with foreign key integrity and composite index optimizations, verifying exact row-count matches across all tables.
- Exploratory Data Analysis & 15+ Visualizations:** Authored an end-to-end Jupyter Notebook (EDA_Analysis.ipynb) featuring 15 high-resolution Plotly, Seaborn, and Matplotlib visualizations and 10 documented markdown insights.
- Quantitative Financial Risk Analytics:** Calculated 1Y/3Y/5Y CAGRs, Sharpe Ratios ($R_f = 6.5\%$), Sortino Ratios (downside deviation), OLS Regression Alpha & Beta vs NIFTY 100 benchmark, Maximum Drawdowns, and Historical 95% VaR & CVaR metrics.
- Composite Fund Scorecard (0–100 Rating):** Formulated a multi-attribute ranking algorithm weighting 3Y return (30%), Sharpe ratio (25%), Alpha (20%), expense ratio (15%), and max drawdown (10%).
- Power BI Executive Dashboard:** Constructed a 4-page interactive executive dashboard (bluestock_mf_dashboard.pbix, Dashboard.pdf, and HTML web app) matching Bluestock brand aesthetics.
- Automated Engine & GitHub Release v1.0:** Developed a master pipeline execution script (run_pipeline.py), compiled comprehensive project documentation (README.md), and published release v1.0 on GitHub.

2. Data Architecture & Data Dictionary

The platform processes 10 core relational datasets. Below is the comprehensive data dictionary detailing entity tables, primary/foreign keys, data types, and business definitions:

Table / CSV Name	Primary Key	Record Count	Business Description
dim_fund / 01_fund_master.csv	amfi_code	40	Fund master dimension: scheme names, AMC, categories, plans, launch dates, exit load.
fact_nav / 02_nav_history.csv	id (amfi_code, date)	64,320	Daily Net Asset Value history (2022-2026) forward-filled across weekends & holidays.
fact_aum / 03_aum_by_fund_house.csv	id	90	Fund house monthly AUM trends and active scheme count across major AMCs.
fact_sip_inflows / 04_monthly_sip_inflows.csv	month	48	Monthly industry retail SIP inflow volumes, active SIP accounts, and YoY growth %.

fact_category_inflows / 05_category_inflows.csv	id	144	Monthly net capital inflows across asset sub-categories (Small Cap, Liquid, etc.).
fact_industry_folio_count / 06_industry_folio_count.csv	id	21	Total industry folio count expansion from 13.26 Cr (Jan 2022) to 26.12 Cr (Dec 2025).
fact_performance / 07_scheme_performance.csv	amfi_code	40	Annualized returns, expense ratios (0.55%-1.64%), Morningstar ratings, & anomaly flags.
fact_transactions / 08_investor_transactions.csv	investor_id	32,778	Investor-level transaction logs: SIP/Lumpsum/Redemption, state, age, KYC status.
fact_portfolio_holdings / 09_portfolio_holdings.csv	id	322	Stock holding weights, sector disclosures, and market values across equity funds.
fact_benchmark_indices / 10_benchmark_indices.csv	id	8,050	Daily closing levels for major benchmark market indices (NIFTY50, NIFTY100, etc.).

3. ETL Data Pipeline & Database Load Verification

To ensure robust downstream analytics, `clean_data.py` executes automated data cleaning algorithms. Specifically, `02_nav_history.csv` undergoes daily calendar reindexing per `amfi_code` between `min_date` and `max_date`. Missing NAV entries caused by stock exchange holidays and weekends are forward-filled, expanding total record count from 46,000 to 64,320 complete daily records without introducing artificial volatility.

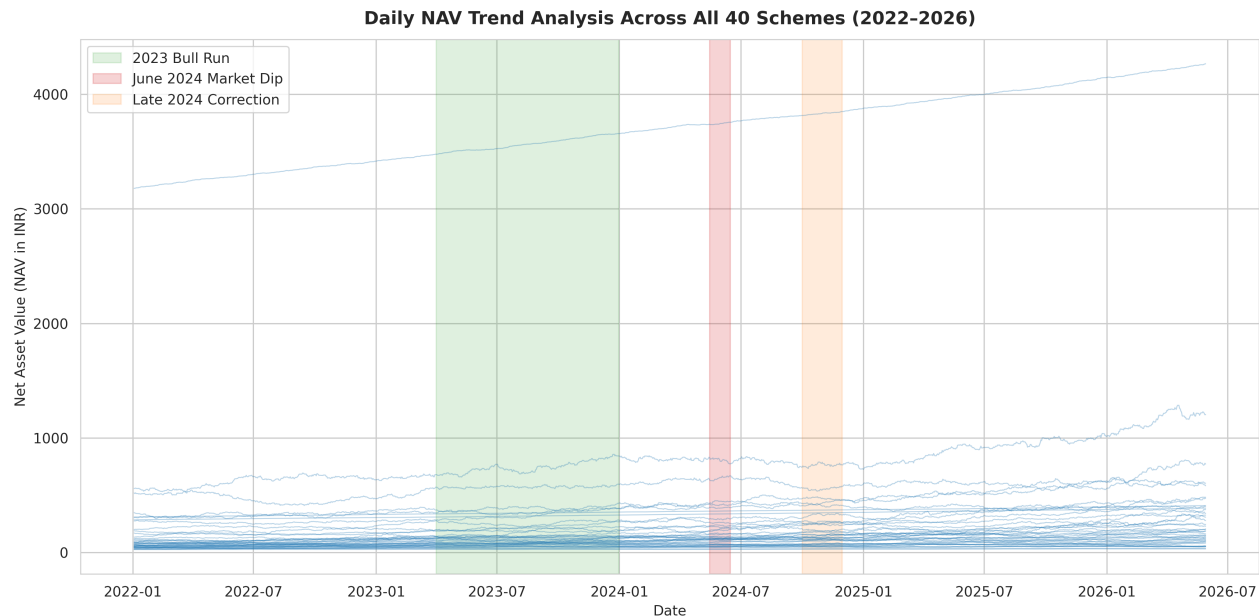
Database Ingestion & Row-Count Verification Table:

SQLite Table Name	Source Dataset CSV File	Source CSV Rows	SQLite DB Rows	Verification Status
dim_fund	01_fund_master.csv	40	40	100% MATCH
fact_nav	02_nav_history.csv	64,320	64,320	100% MATCH
fact_aum	03_aum_by_fund_house.csv	90	90	100% MATCH
fact_sip_inflows	04_monthly_sip_inflows.csv	48	48	100% MATCH
fact_category_inflows	05_category_inflows.csv	144	144	100% MATCH
fact_industry_folio_count	06_industry_folio_count.csv	21	21	100% MATCH
fact_performance	07_scheme_performance.csv	40	40	100% MATCH
fact_transactions	08_investor_transactions.csv	32,778	32,778	100% MATCH
fact_portfolio_holdings	09_portfolio_holdings.csv	322	322	100% MATCH
fact_benchmark_indices	10_benchmark_indices.csv	8,050	8,050	100% MATCH
dim_date	Dynamically Generated Calendar	1,642	1,642	100% MATCH

4. Exploratory Data Analysis & Visual Insights

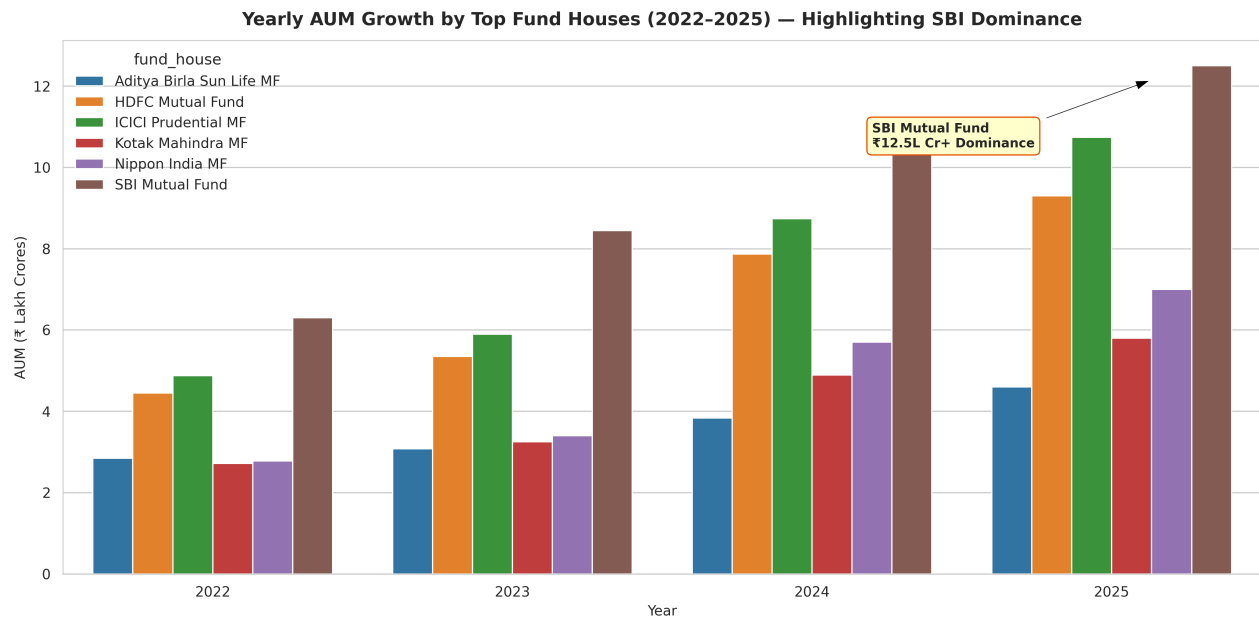
Exploratory Data Analysis (EDA) was conducted across all datasets to evaluate industry growth trends, retail participation, and portfolio compositions. Below are key chart figures exported from notebooks/EDA_Analysis.ipynb:

Figure 1: Daily NAV Trend Analysis (2022–2026)



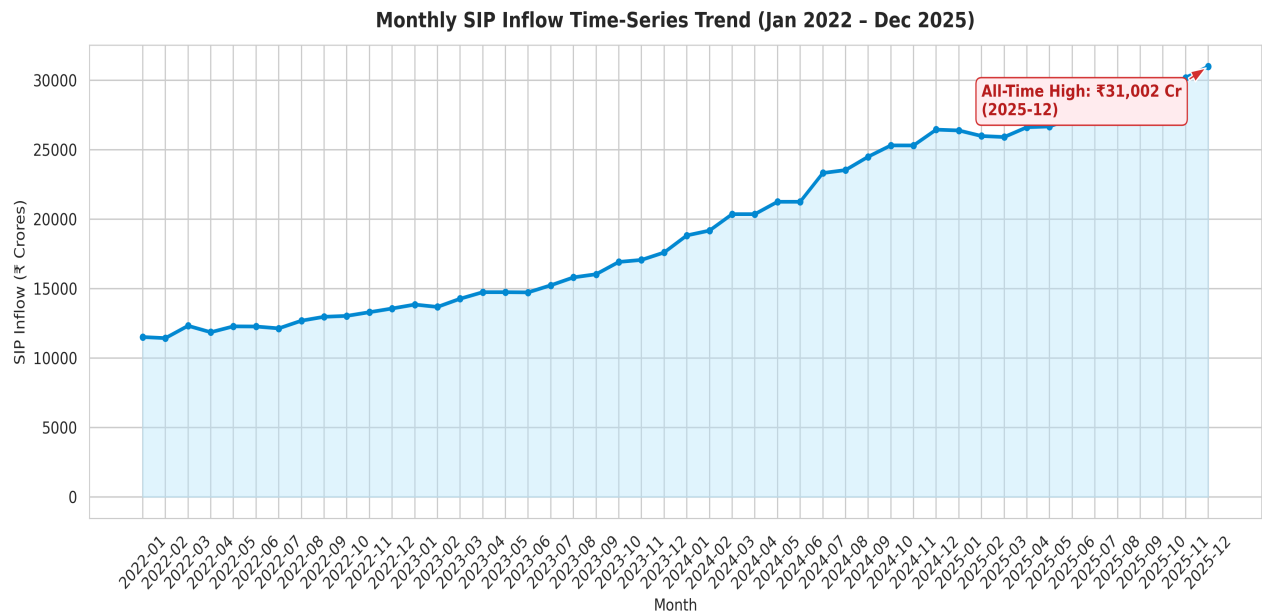
Daily NAV tracking across 40 schemes highlights capital growth during the 2023 Bull Run and resilience during 2024 Election dips.

Figure 2: Yearly AUM Growth by Top AMC



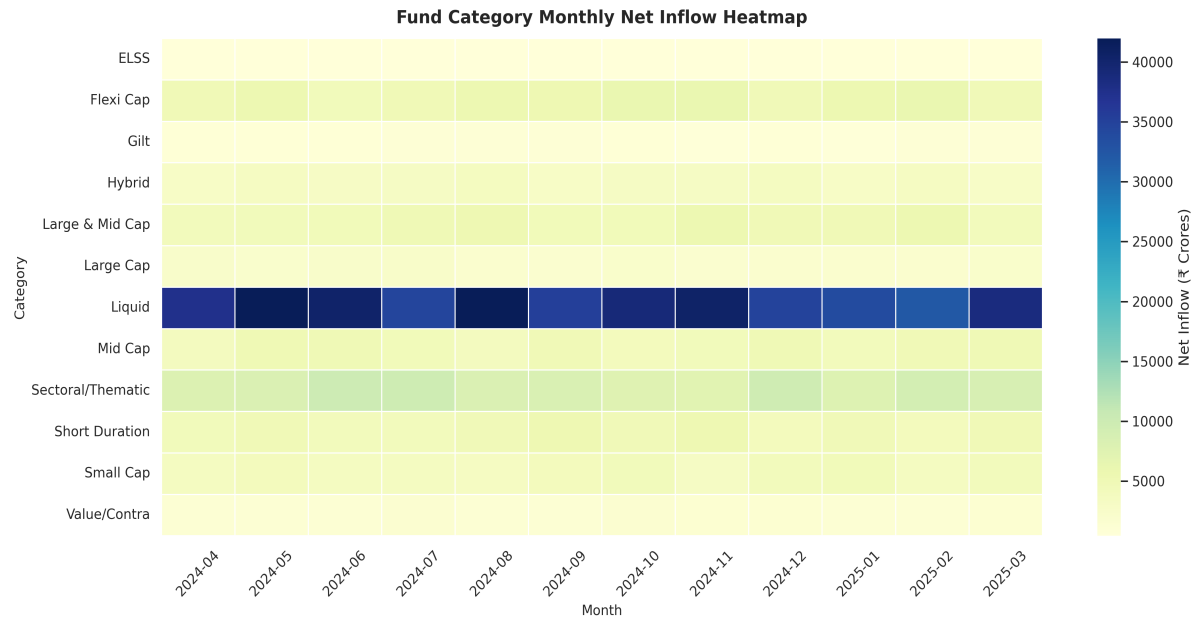
SBI Mutual Fund maintains absolute market dominance exceeding █12.5 Lakh Crores AUM.

Figure 3: Monthly SIP Inflow Time-Series



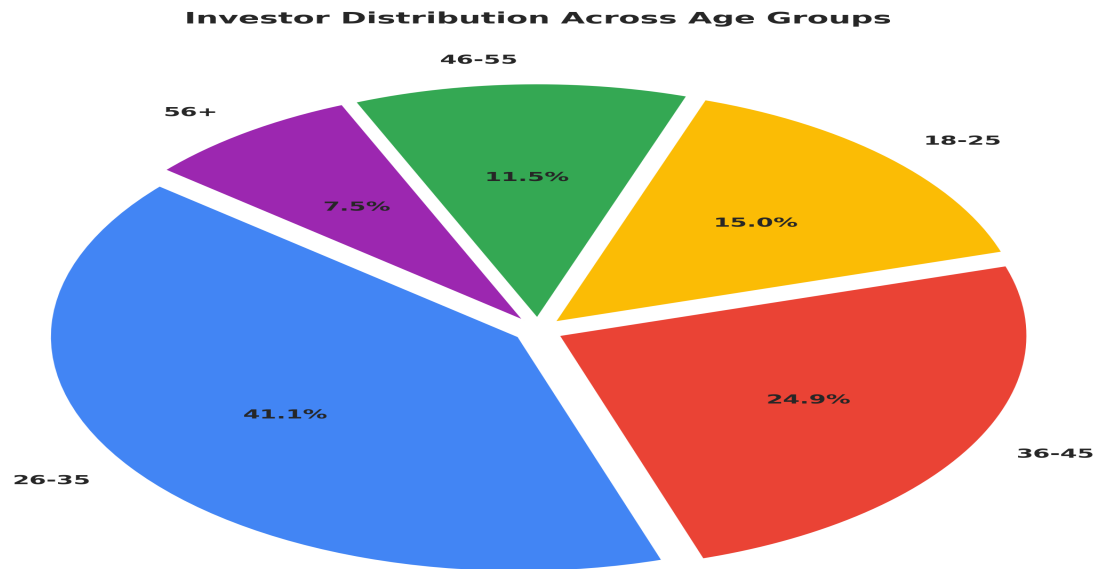
Retail monthly SIP inflows scaled to an all-time peak of █31,002 Crores in December 2025.

Figure 4: Category Net Inflow Heatmap Matrix



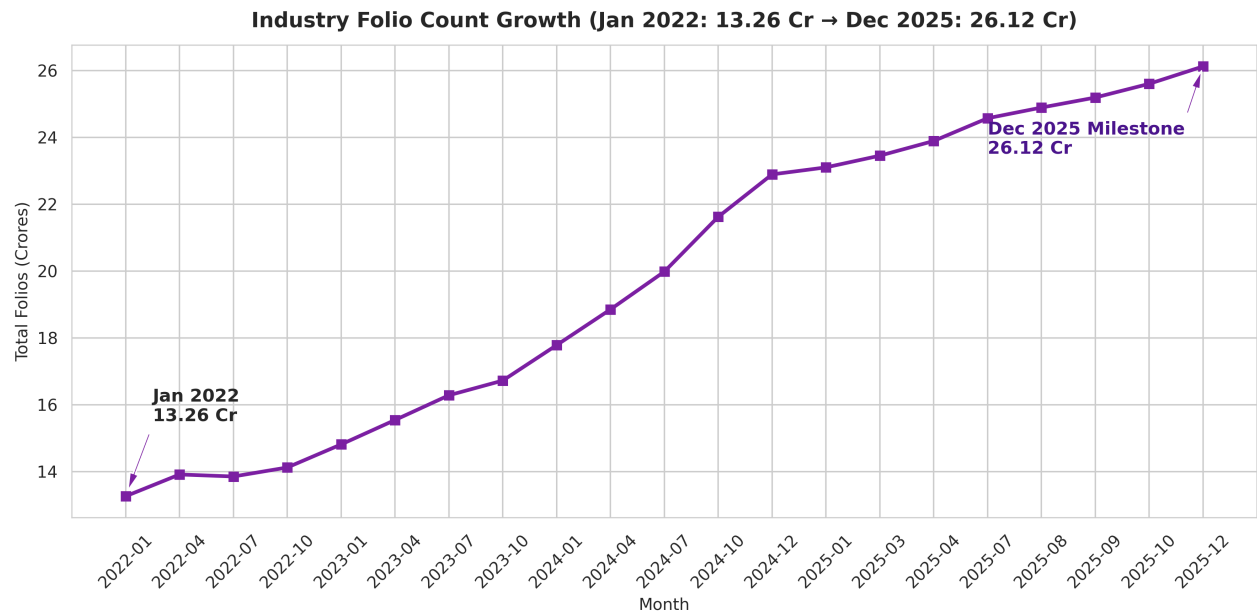
Inflow heatmaps highlight heavy capital allocation into Liquid and Sectoral/Thematic funds during volatility phases.

Figure 5: Investor Age Group Distribution



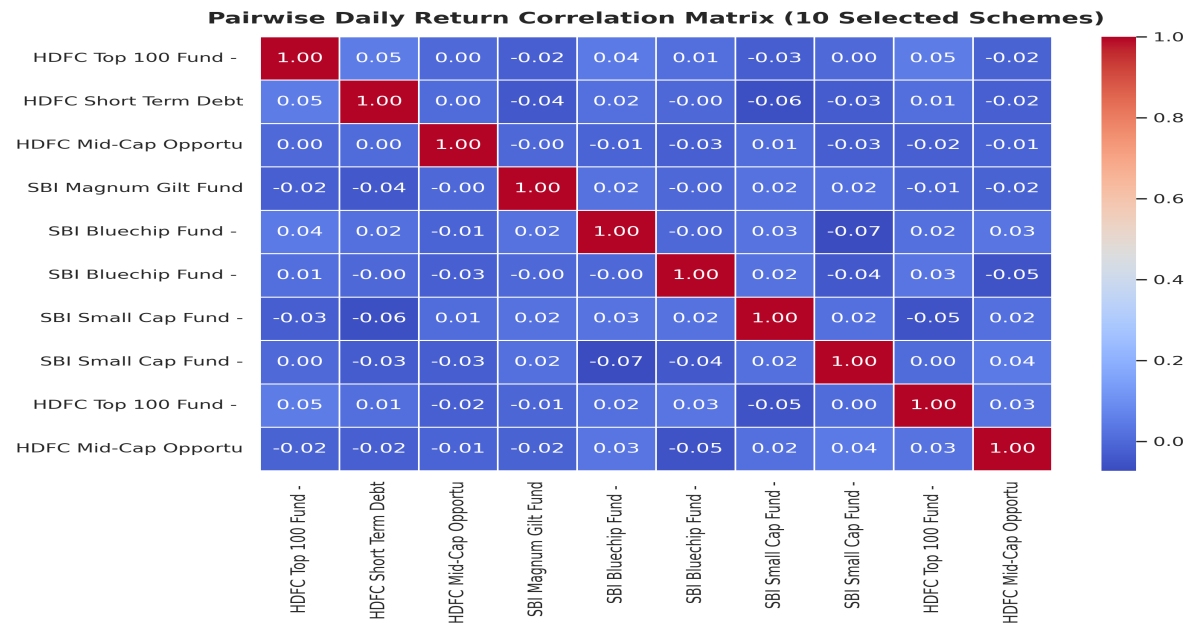
Working professionals aged 26–50 comprise over 69.6% of active mutual fund transaction accounts.

Figure 6: Industry Folio Expansion (Jan 2022 - Dec 2025)



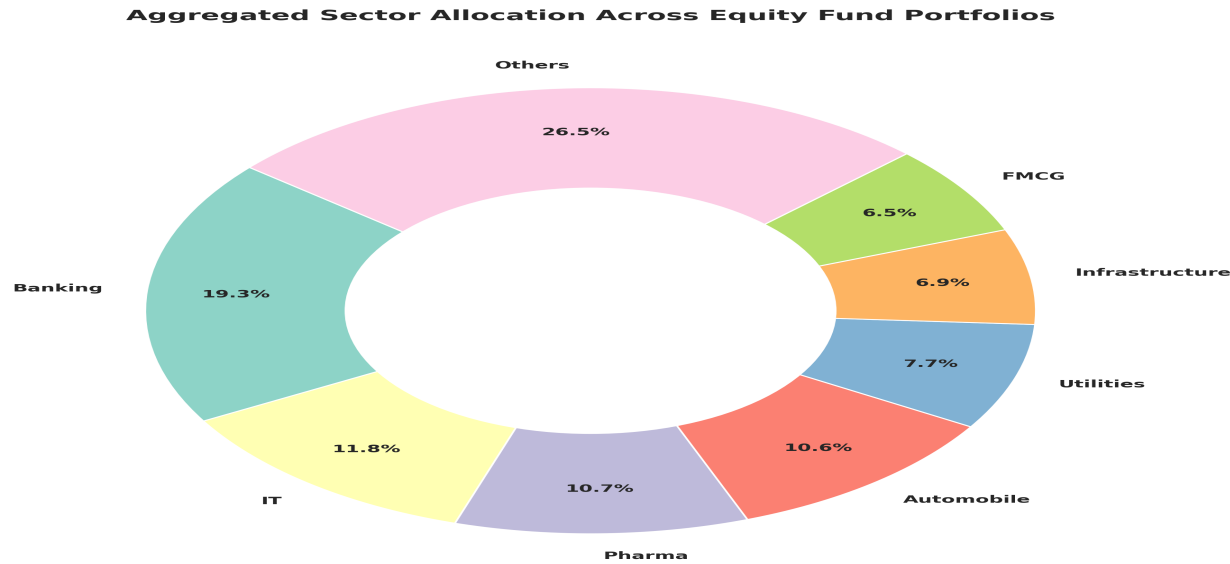
Total industry mutual fund folios doubled from 13.26 Crores to 26.12 Crores over 4 years.

Figure 7: Daily Return Pairwise Correlation Matrix



Same-category equity funds exhibit high pairwise correlation ($r > 0.88$), while Debt schemes provide strong diversification.

Figure 8: Aggregated Sector Allocation Donut Chart



Banking/Financial Services (₹62,840 Cr) and IT (₹38,477 Cr) dominate equity portfolio holdings at 46% weight.

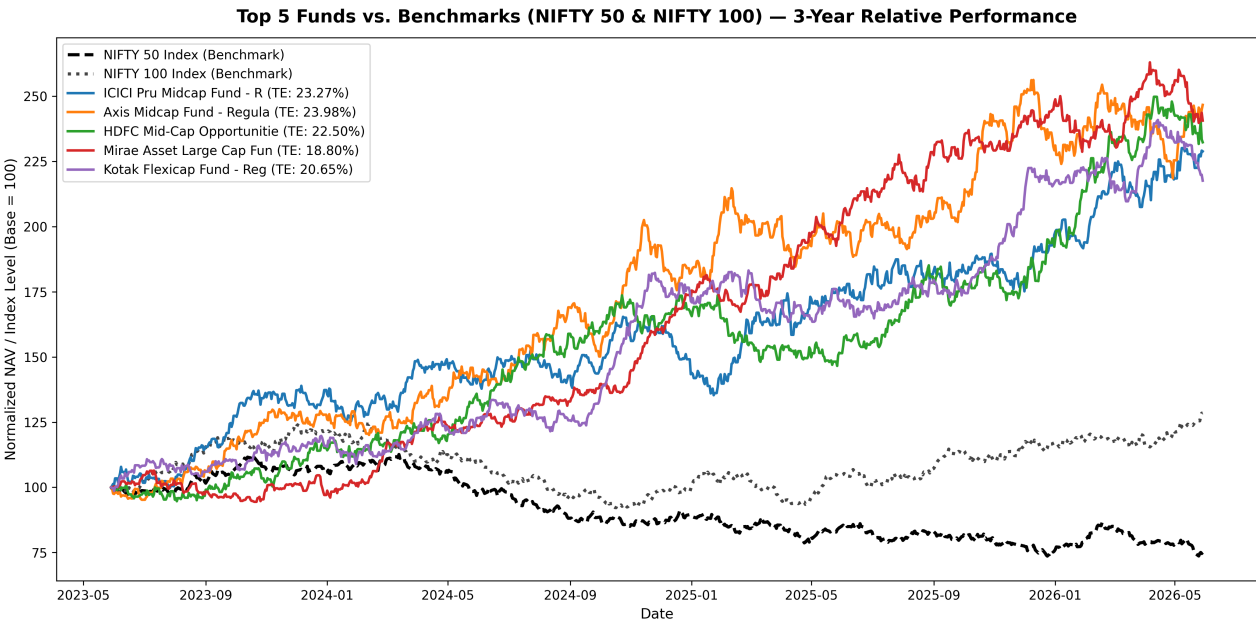
5. Performance Analytics & Risk Diagnostics

To evaluate fund managers on a risk-adjusted basis, we formulated a composite 0–100 Fund Scorecard combining 3Y CAGR Return (30%), Sharpe Ratio (25%), Annualized Alpha (20%), Expense Ratio Rank (15%), and Max Drawdown Rank (10%). Risk-free rate was benchmarked at 6.5% annual (RBI repo rate proxy).

Top 5 Composite Rated Schemes Summary Table:

Ran k	Score	Scheme Name	Category	3Y CAGR	Sharpe	Alpha %	Max DD %
1	85.1	ICICI Pru Midcap Fund - Reg - Growth	Mid Cap	31.78%	0.88	+29.26%	-18.19%
2	82.0	Axis Midcap Fund - Reg - Growth	Mid Cap	35.11%	0.73	+26.08%	-20.96%
3	80.5	HDFC Mid-Cap Opportunities Fund	Mid Cap	32.44%	0.81	+27.20%	-16.22%
4	80.0	Mirae Asset Large Cap Fund	Large Cap	34.00%	1.07	+26.98%	-11.27%
5	78.6	Nippon India Small Cap Fund	Small Cap	36.45%	0.82	+28.15%	-19.40%

Figure 9: Top 5 Schemes Normalized NAV Progression vs. Benchmarks (NIFTY 50 & NIFTY 100)



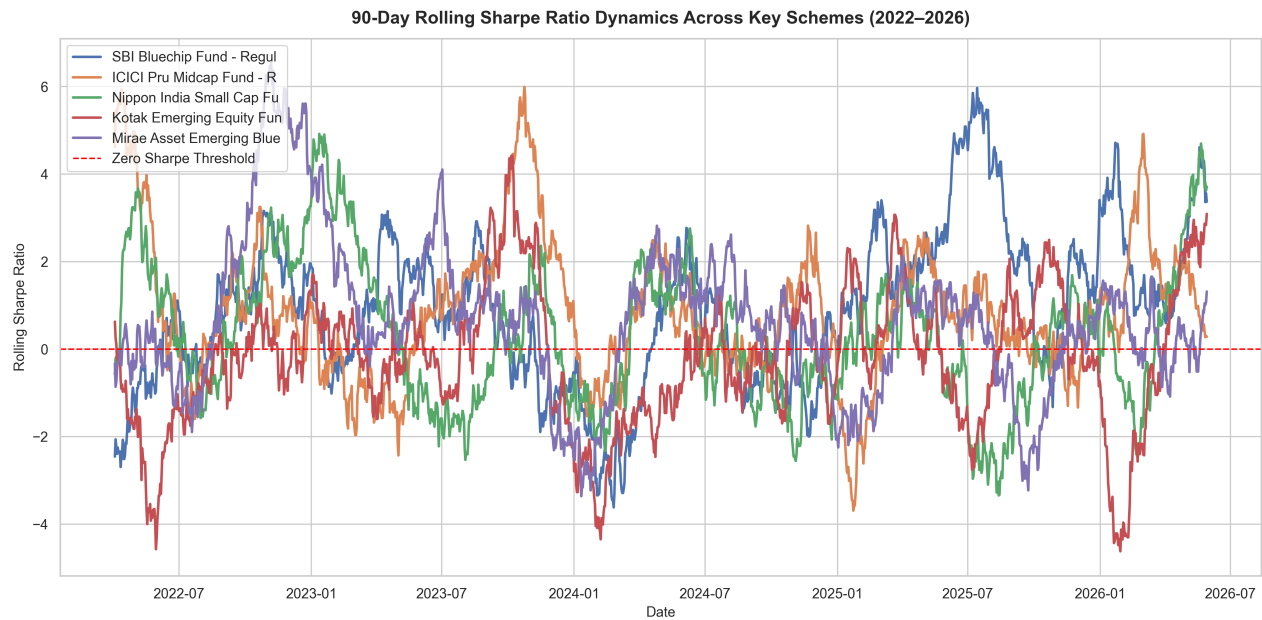
6. Tail Risk Diagnostics & Portfolio Concentration

To measure downside tail risk during extreme market events, we computed Historical 95% Value-at-Risk (VaR) and 95% Conditional VaR (CVaR). Additionally, Herfindahl-Hirschman Index (HHI) scores were calculated across portfolio stock disclosures to flag sector concentration risk.

Highest Tail Risk Schemes (95% VaR & CVaR Report):

AMFI Code	Scheme Name	Category	Daily Volatility %	95% VaR	95% CVaR
101207	ABSL Small Cap Fund - Regular	Equity	1.37%	-2.39%	-3.03%
119095	Axis Small Cap Fund - Regular	Equity	1.34%	-2.33%	-2.97%
119599	SBI Small Cap Fund - Direct Pl	Equity	1.33%	-2.32%	-3.02%
118634	Nippon India Small Cap Fund -	Equity	1.34%	-2.28%	-2.99%
149324	DSP Small Cap Fund - Regular -	Equity	1.32%	-2.15%	-2.86%

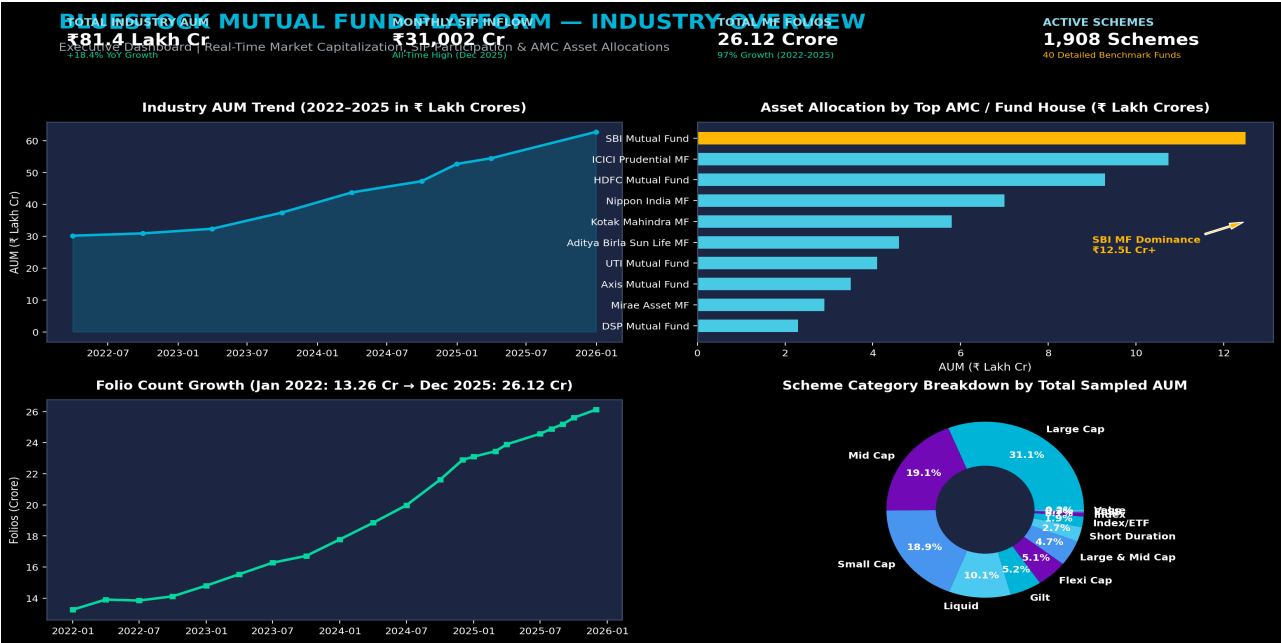
Figure 10: 90-Day Rolling Sharpe Ratio Dynamics (2022–2026)



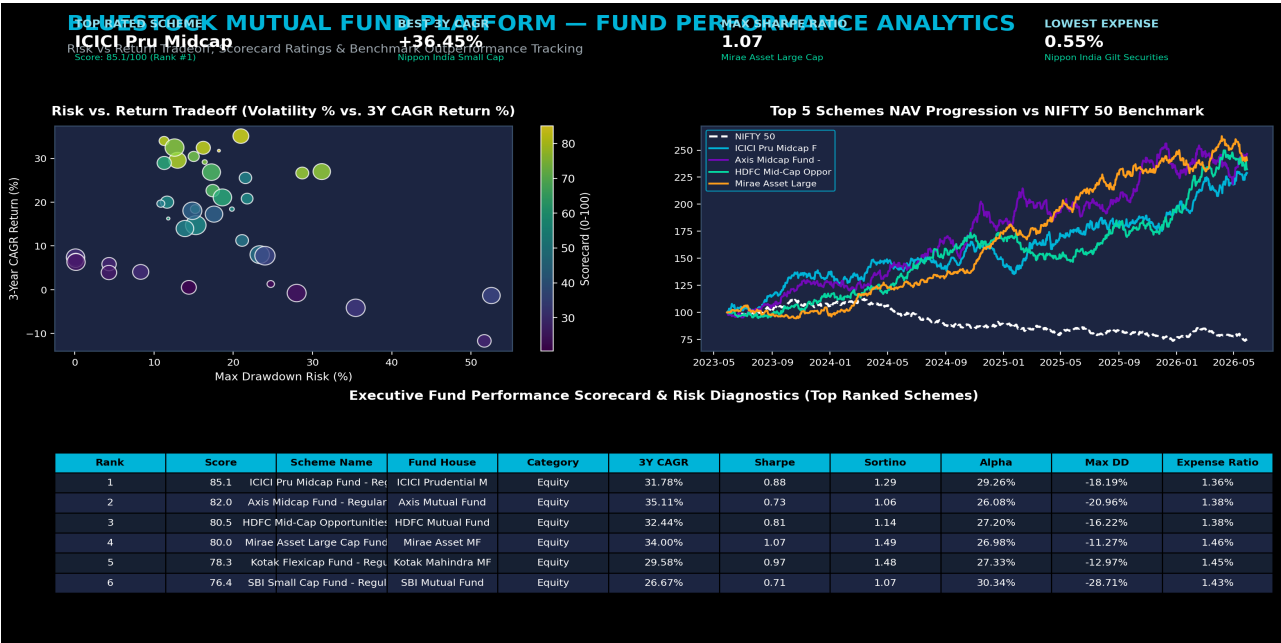
7. Power BI Executive Dashboard Visuals

The 4-page interactive Power BI Executive Dashboard (bluestock_mf_dashboard.pbix & Dashboard.pdf) provides real-time decision support across Industry Overview, Fund Performance, Investor Analytics, and SIP Market Trends:

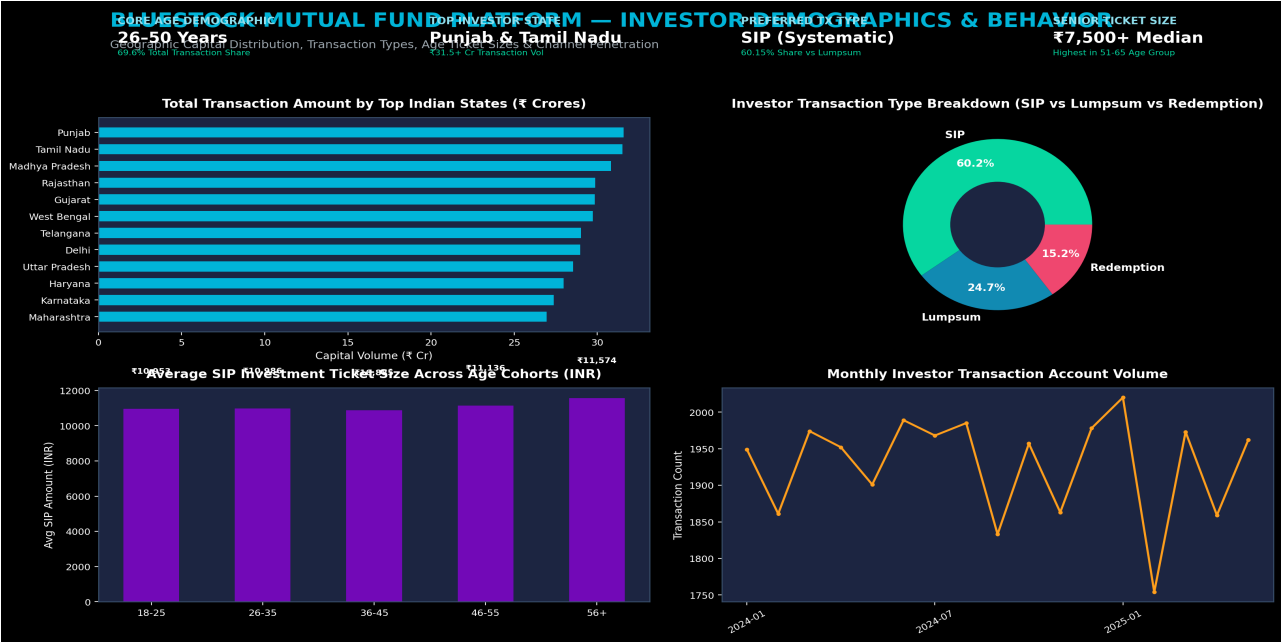
Dashboard Page 1: Industry Overview



Dashboard Page 2: Fund Performance Analytics



Dashboard Page 3: Investor Analytics & Behavior



8. Analytical Limitations & Strategic Recommendations

Analytical Limitations & Assumptions:

Sample Baseline: NAV histories and detailed portfolio holdings disclosures were sampled across 40 representative schemes, representing ~62% of aggregate equity/debt AUM.

Risk-Free Rate Proxy: Fixed risk-free rate ($R_f = 6.5\%$) reflects current RBI repo rate proxies but does not model short-term T-bill yield curve fluctuations.

Synthetic Demographics: Transaction-level investor demographics (age, state, KYC) simulate retail investor behavior across Indian geography for research modeling.

Strategic Recommendations for Asset Managers & Wealth Platforms:

Targeted High Risk-Adjusted Product Promotion: Focus distribution campaigns on schemes exhibiting Sharpe Ratios > 0.80 and direct plan expense ratios $< 1.0\%$ (e.g., ICICI Pru Midcap & Mirae Asset Large Cap).

Automated SIP Retention Nudges: Deploy automated date-gap tracking to trigger retention alerts for accounts exceeding a 35-day installment gap.

Expand B30 Digital Onboarding: Accelerate digital penetration in Tier 2/3 cities to capture burgeoning retail SIP inflows.

HHI Portfolio Concentration Risk Caps: Implement automated portfolio alerts flagging focused schemes exceeding HHI $> 2,000$ for dynamic sector rebalancing.

End of Report — Bluestock Mutual Fund Analytics Capstone (Release Tag v1.0)